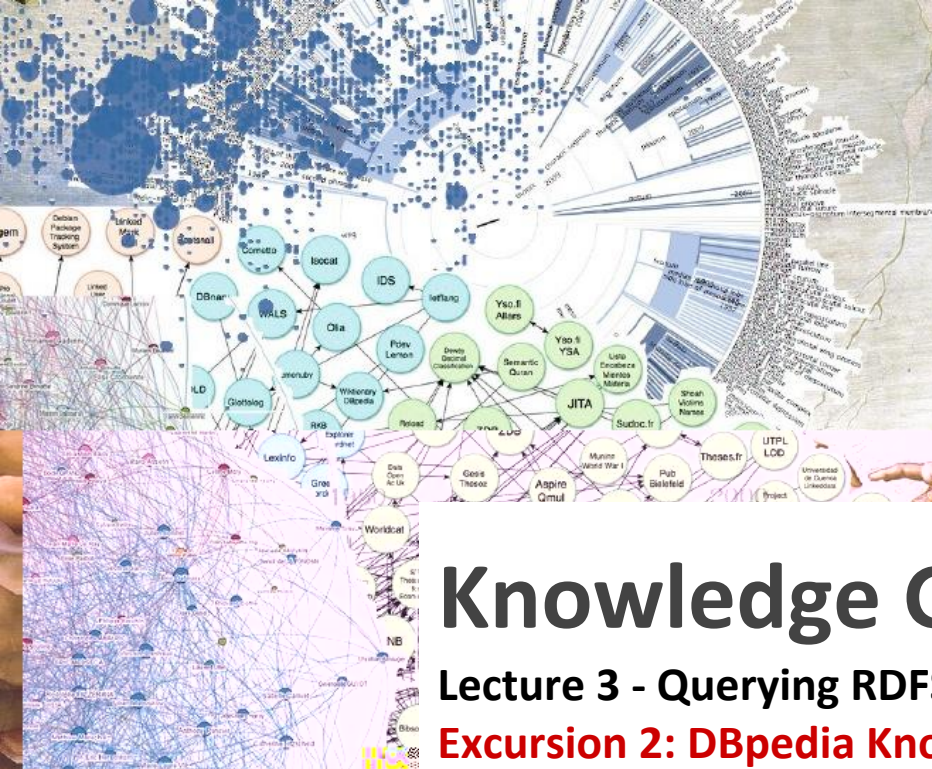




Knowledge Graphs

Lecture 3 - Querying RDFS with SPARQL

Excursion 2: DBpedia Knowledge Graph

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AIFB - Karlsruhe Institute of Technology

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Knowledge Graphs

Lecture 3: Querying RDF(S) with SPARQL

3.1 How to Query RDF(S)

Excursion 2: DBpedia Knowledge Graph

Excursion 3: Wikidata Knowledge Graph

3.2 Complex Queries with SPARQL

3.3 More Complex SPARQL Queries

3.4 SPARQL Subqueries and Property Paths

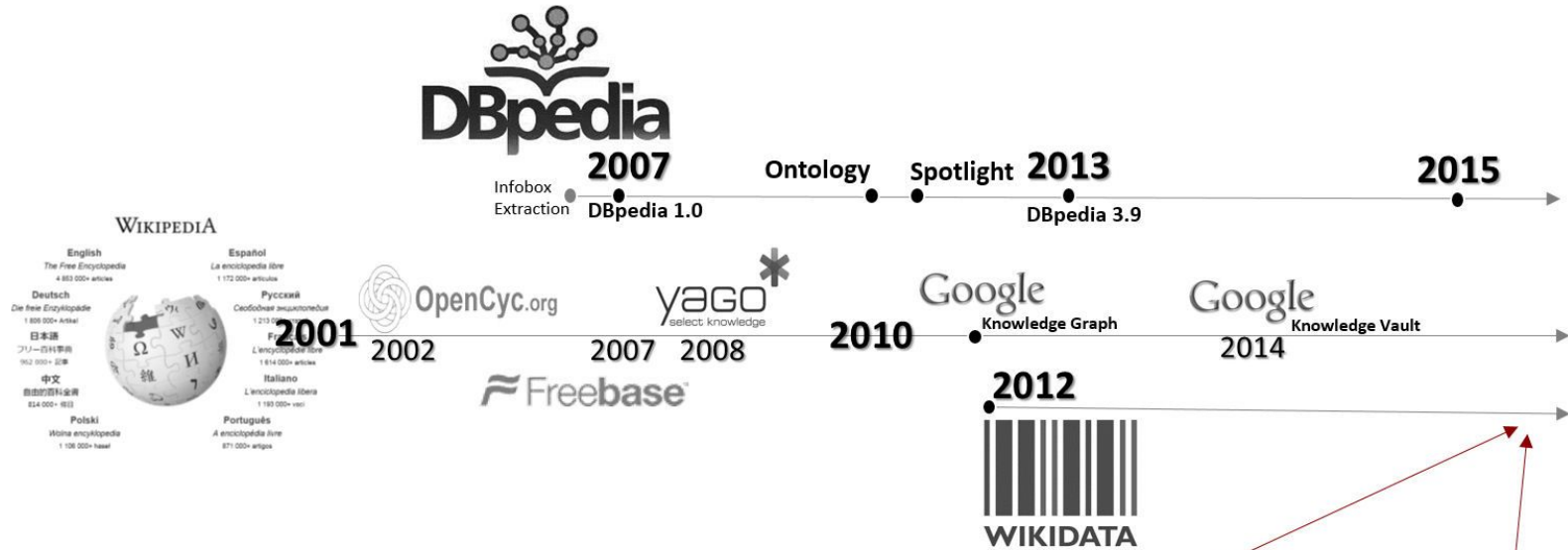
3.5 RDF Databases

3.6 SPARQL is more than a Query Language



- a central hub in the Web of Data

Knowledge Bases Timeline



DBpedia Knowledge Graph

English version of the DBpedia Knowledge Graph (as by January 2020)

- describes 6.6 million things,
- of which 5.5 million are classified in a consistent [ontology](#)
- including 1.5 million persons,
- 840,000 places (including 513,000 populated places),
- 496,000 creative works
 - including 139,000 music albums,
 - 111,000 films and
 - 21,000 video games,
- 286,000 organizations
 - including 70,000 companies and 55,000 educational institutions,
- 306,000 species and
- 6,000 diseases.

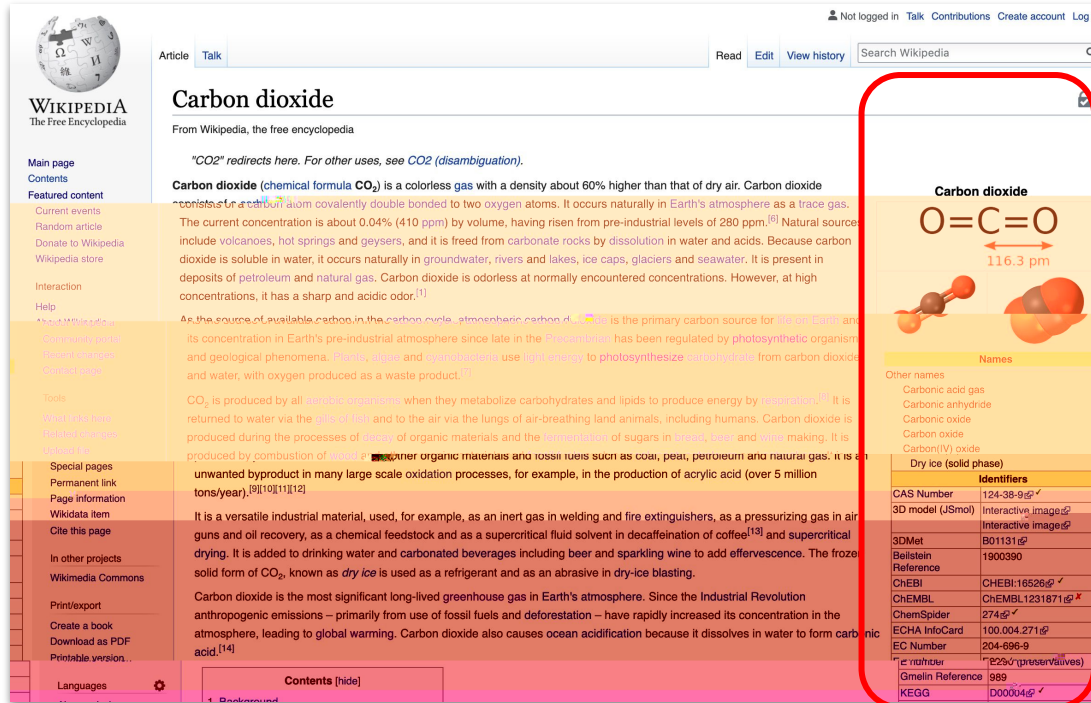


<https://wiki.dbpedia.org/develop/datasets>

3. Querying RDF(S) with SPARQL / Excursion 2: DBpedia Knowledge Graph

From Wikipedia to DBpedia

http://en.wikipedia.org/wiki/Carbon_dioxide

Article Talk

Read Edit View history Search Wikipedia

Carbon dioxide

From Wikipedia, the free encyclopedia

"CO2" redirects here. For other uses, see *CO2 (disambiguation)*.

Carbon dioxide (chemical formula **CO₂**) is a colorless gas with a density about 60% higher than that of dry air. Carbon dioxide consists of a carbon atom covalently double bonded to two oxygen atoms. It occurs naturally in Earth's atmosphere as a trace gas. The current concentration is about 0.04% (410 ppm) by volume, having risen from pre-industrial levels of 280 ppm.^[6] Natural sources include volcanoes, hot springs and geysers, and it is freed from carbonate rocks by dissolution in water and acids. Because carbon dioxide is soluble in water, it occurs naturally in groundwater, rivers and lakes, ice caps, glaciers and seawater. It is present in deposits of petroleum and natural gas. Carbon dioxide is odorless at normally encountered concentrations. However, at high concentrations, it has a sharp and acidic odor.^[1]

As the source of available carbon in the carbon cycle, atmospheric carbon dioxide is the primary carbon source for photosynthesis and its concentration in Earth's pre-industrial atmosphere since late in the Pleistocene has been regulated by photosynthetic organisms and geological phenomena. Plants, algae and cyanobacteria use light energy to photosynthesize carbohydrates from carbon dioxide and water, with oxygen produced as a waste product.^[2]

CO₂ is produced by all aerobic organisms when they metabolize carbohydrates and lipids to produce energy by respiration.^[6] It is returned to water via the gills of fish and to the air via the lungs of air-breathing land animals, including humans. Carbon dioxide is produced during the processes of decay of organic materials and the fermentation of sugars in bread, beer and wine making. It is produced by combustion of wood and other organic materials and fossil fuels such as coal, peat, petroleum and natural gas. It is an unwanted byproduct in many large scale oxidation processes, for example, in the production of acrylic acid (over 5 million tons/year).^{[9][10][11][12]}

It is a versatile industrial material, used, for example, as an inert gas in welding and fire extinguishers, as a pressurizing gas in air guns and oil recovery, as a chemical feedstock and as a supercritical fluid solvent in decaffeination of coffee^[13] and supercritical drying. It is added to drinking water and carbonated beverages including beer and sparkling wine to add effervescence. The frozen solid form of CO₂, known as *dry ice* is used as a refrigerant and as an abrasive in dry-ice blasting.

Carbon dioxide is the most significant long-lived greenhouse gas in Earth's atmosphere. Since the Industrial Revolution anthropogenic emissions – primarily from use of fossil fuels and deforestation – have rapidly increased its concentration in the atmosphere, leading to global warming. Carbon dioxide also causes ocean acidification because it dissolves in water to form carbonic acid.^[14]

Carbon dioxide

O=C=O

116.3 pm

Names

Other names

- Carbonic acid gas
- Carbonic anhydride
- Carbonic oxide
- Carbon oxide
- Carbon(IV) oxide

Dry ice (solid phase)

Identifiers	
CAS Number	124-38-9 ⓘ
3D model (JSmol)	Interactive image ⓘ
3D model	B01131 ⓘ
Beilstein Reference	1900390
ChEBI	ChEBI:16526 ⓘ
ChEMBL	ChEMBL1231871 ⓘ
ChemSpider	274 ⓘ
ECHA InfoCard	100.004.271 ⓘ
EC Number	204-696-9
ECHA InfoCard	100.004.271 ⓘ
Gmelin Reference	989
KEGG	D00004 ⓘ



http://dbpedia.org/resource/Carbon_dioxide

From Wikipedia to DBpedia

http://dbpedia.org/resource/Carbon_dioxide



The screenshot shows the DBpedia web interface for the entity 'Carbon dioxide'. The header includes the DBpedia logo, navigation links for 'Browse using' and 'Formats', and links to a 'Faceted Browser' and 'Sparql Endpoint'. The main heading is 'About: Carbon dioxide'. Below this, a metadata line states: 'An Entity of Type : chemical compound, from Named Graph : http://dbpedia.org, within Data Space : dbpedia.org'. The main content area is mostly obscured by a large green rectangle, with only the beginning of a paragraph visible: 'Carbon dioxide (chemical formula CO₂) is a colorless and odorless gas vital to life on Earth'.

DBpedia Naming Conventions

https://en.wikipedia.org/wiki/Carbon_dioxide



WIKIPEDIA
The Free Encyclopedia

http://dbpedia.org/resource/Carbon_dioxide

Entity Identifier

http://dbpedia.org/page/Carbon_dioxide

HTML version

http://dbpedia.org/data/Carbon_dioxide

RDF/XML version



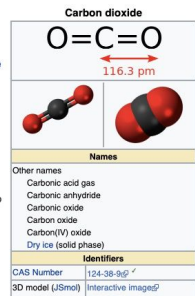
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2

2nanana9

DBpedia Category System - DBpedia Ontology Classes

dbr:Carbon_dioxide



rdf:type

dbo:Chemical_compound

rdfs:subClassOf

dbo:Chemical_substance

rdfs:subClassOf

owl:Thing

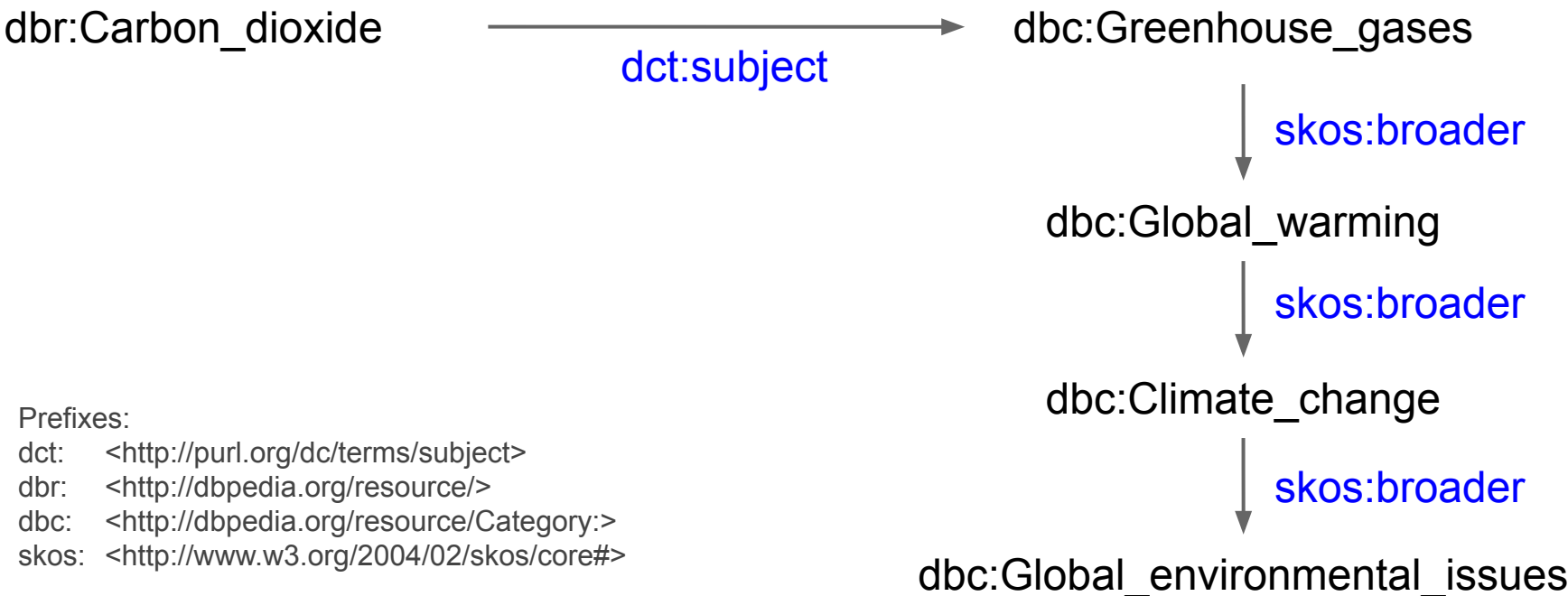
Prefixes:

rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
 dbr: <http://dbpedia.org/resource/>
 dbo: <http://dbpedia.org/ontology/>
 owl: <http://www.w3.org/2002/07/owl#>
 rdfs: <http://www.w3.org/2000/01/rdf-schema#>

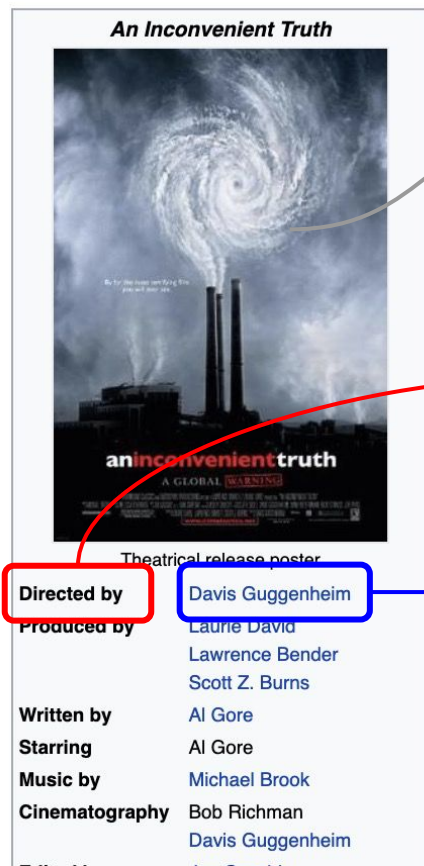
DBpedia Category System - Wikipedia Categories

https://en.wikipedia.org/wiki/Carbon_dioxide

Categories: Carbon dioxide | Acid anhydrides | Acidic oxides | Coolants | Fire suppression agents | Greenhouse gases | Household chemicals | Inorganic solvents | Laser gain media | Nuclear reactor coolants | Oxocarbons | Propellants | Refrigerants | Gaseous signaling molecules | Heterocumulenes | E-number additives



DBpedia Infobox Extraction - Infobox Properties



http://dbpedia.org/resource/An_Inconvenient_Truth

<http://dbpedia.org/property/director>

http://dbpedia.org/resource/Davis_Guggenheim



Next Lecture:

Excursion 3: Wikidata Knowledge Graph

Picture References:

- [1] DBpedia logo, [wiki.dbpedia.org](https://commons.wikimedia.org/wiki/File:DBpediaLogo.svg), DBpedia Team [Public Domain]
<https://commons.wikimedia.org/wiki/File:DBpediaLogo.svg>
- [2] The Linked Open Data Cloud, lod-cloud.net, [CC-BY]
<https://lod-cloud.net/clouds/lod-cloud.svg>
- [3] Andrea Huang, A preliminary study on Wikipedia Dbpedia and Wikidata, SlideShare [CC_BY_SA 4.0]
<https://www.slideshare.net/andreasinica/a-preliminary-study-on-wikipedia-dbpedia-and-wikidata>
- [4] Wikidata Logo, Planemad, [Public Domain]
<https://commons.wikimedia.org/wiki/File:Wikidata-logo-en.svg>